

SYSTEMS SIMULATION

Systems Analysis & Design

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UNIVERSIDAD DISTRITAL
FRANCISCO JOSÉ DE CALDAS

Outline

1 Basic Concepts

2 Cellular Automata

3 Digital Twins



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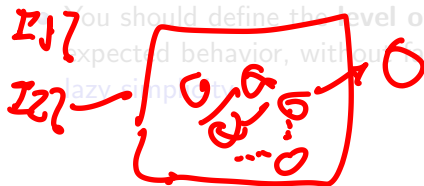
What is a simulation?

- **Simulations** are a type of real-world representation, but rarely exactly like the real world.
- Sometimes you need to **test** or **experiment** with expensive use cases. Simulations let you **play** with different inputs, conditions, hyperparameter optimizations.
- Also, there are dangerous or hard-to-reach **scenarios** where **simulations** become the best option.
- You should define the **level of detail** well enough to represent the expected behavior, without failing due to either **high complexity** or **lazy simplicity**.



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 - Also, there are dangerous or hard-to-reach scenarios where simulations become the best option.
 - You should define the level of detail well enough to represent the expected behavior without falling due to either high complexity or lazy simplicity.
- Handwritten red notes:*
Fly a plane
drive fg
(An arrow points from the text "You should define the level of detail" to the handwritten notes)



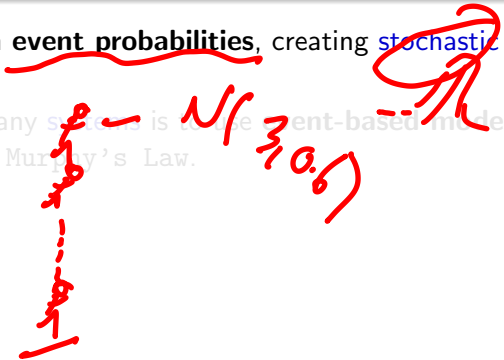
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Events and Stochastic Processes

- It is typical to play with **event probabilities**, creating **stochastic behaviors**.
- One way to **simulate** many **systems** is to use **event-based models**.
Embrace the chaos and Murphy's Law.



A
4(3.5)

B
7

C
Trig. (m)

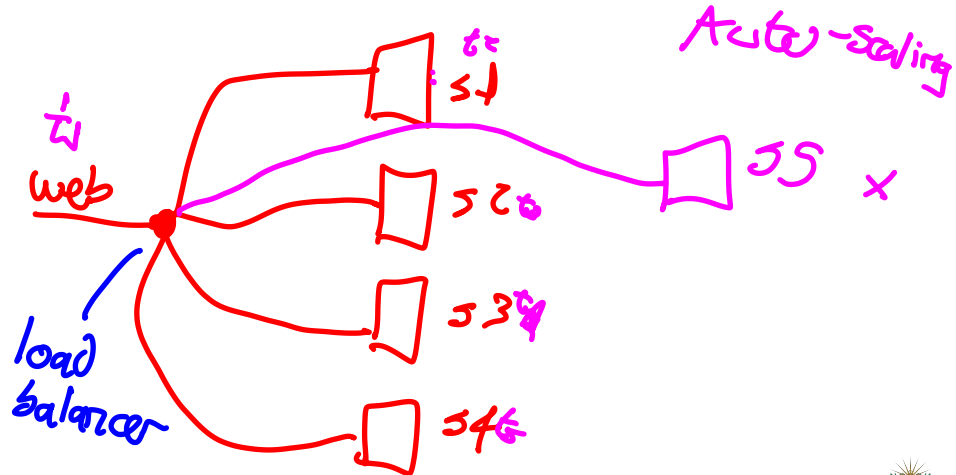


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Case Study: Load Balancer in a Cloud Architecture



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Cellular Automata

- Cellular Automata are a **discrete model** defined by a **grid** of cells, each with a **state**. *value*
- The **state** of a cell is updated based on the **states of its neighbors**.

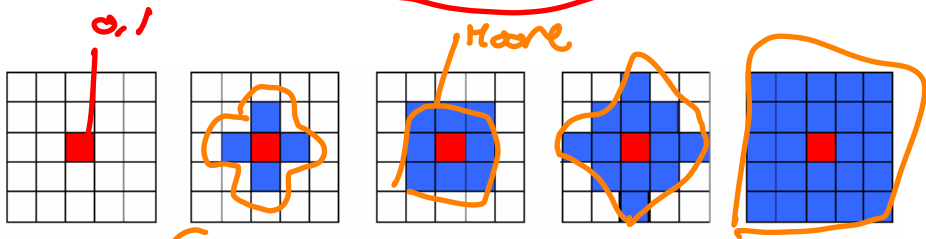


Figure: Cellular Automata Typical Neighborhoods

spatial behaviors



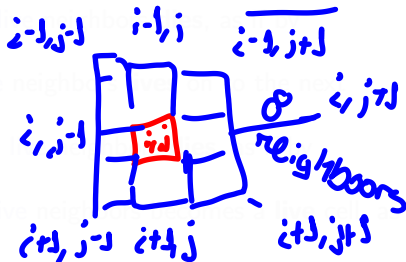
Game of Life

- **Game of Life** is a **cellular automaton** devised by the British mathematician **John Horton Conway** in **1970**.
- It is a **zero-player game**, meaning that its **evolution** is determined by its **initial state**, requiring no further input.

Artificial
Life

chaotic

□ - 0 dead
J - 1 live



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- Rules:

- A** [
- Any **live** cell with fewer than two live neighbors **dies**, as if by underpopulation.
 - Any **live** cell with two or three live neighbors **lives** on to the next generation.
 - Any **live** cell with more than three live neighbors **dies**, as if by overpopulation. ✓
- B** [
- Any **dead** cell with exactly three live neighbors becomes a **live** cell, as if by reproduction.

if state = 1;

A
else:
B

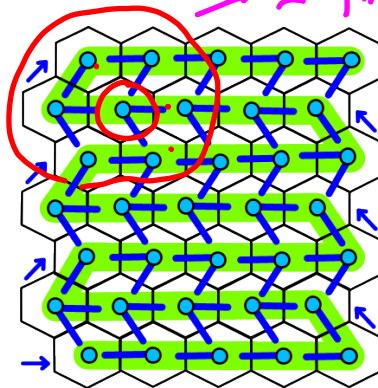
sum = $\sum$ live_neighbors
if $sum < 2$ or $sum > 3$:

A state = 0
B if sum == 3:
state = 1



HoneyComb Cellular Automata

- **HoneyComb Cellular Automata** is a different topology where a cell has six neighbors.
- This representation has different dispersion properties which can be more interesting.



SIR Model

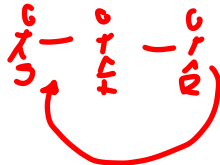
- **SIR Model** is a **compartmental model** used to represent the **transmission** of a **contagious disease**.
- The **model** divides the population into three compartments: **S** for the number of **susceptible individuals**, **I** for the number of **infected individuals**, and **R** for the number of **recovered individuals**.
- The model is defined by the following **differential equations** where β is the **transmission rate** and γ is the **recovery rate**:

$$\frac{dS}{dt} = -\beta \cdot S \cdot I$$

$$\frac{dI}{dt} = \beta \cdot S \cdot I - \gamma \cdot I$$

$$\frac{dR}{dt} = \gamma \cdot I$$

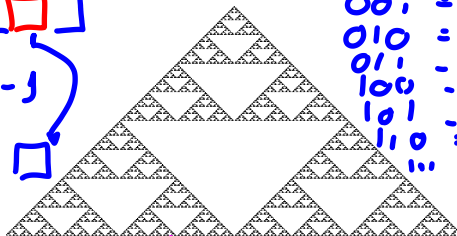
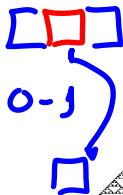
discrete



Chaotic Systems

- **Chaotic Systems** are a class of **dynamical systems** that exhibit **sensitive dependence** on initial conditions.
- This means that the **future behavior** of the system is **highly dependent** on the **initial conditions**.
- The **Lorenz System** is a well-known example of a **chaotic system**.
- Using **cellular automata** to simulate **chaotic systems** is a common practice. Many **fractals** can be created using what are called **chaotic rules**.

2D | 2.1
3D | 3.2
FD | 1.9



Sierpinski

000 = 0
001 = 0
010 = 1
011 = 1
100 = 1
101 = 1
110 = 0
111 = 0

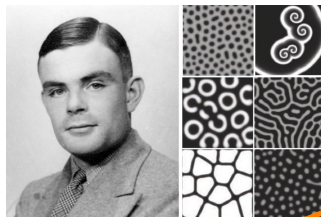
rule 60
0011100



Turing Morphogenesis - Cryptography - AI - Comp. Biology

- **Turing Morphogenesis** is a theory of biological development that explains how patterns form in living organisms.
- The theory is based on the idea that chemical signals can interact to create patterns in living organisms.
- The reaction-diffusion model is a common way to simulate Turing morphogenesis.
 $\text{O O O} \rightarrow \text{reac.} \quad \text{d.f.}$
- The model is defined by a set of reaction and diffusion equations that describe how the chemical signals interact.

equations
inputs
(parameters)



1952 ~ last paper



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What are Digital Twins?

- A **Digital Twin** is a *virtual model* that accurately reflects a physical object or **system**.
- **Digital twins** enable **real-time monitoring**, **simulation**, and **analysis** of the physical counterpart.
- They are widely used in **manufacturing**, **smart cities**, **healthcare**, and **energy systems**.
- By integrating sensor data, **digital twins** help in **predictive maintenance** and **process optimization**.



Applications of Digital Twins

- **Manufacturing:** Monitor equipment performance and **predict failures**.
- **Urban Planning:** Model city infrastructure to **optimize traffic** and resource use.
- **Healthcare:** **Simulate** patient-specific **treatments** and optimize procedures.
- **Energy Systems:** Analyze **grid performance** and integrate renewable energy sources.



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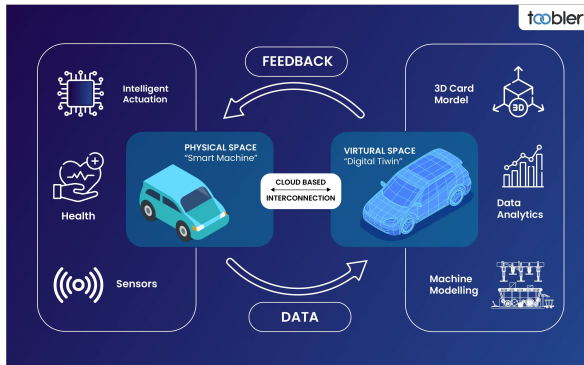
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Benefits of Digital Twins

- Enhanced **decision-making** through **real-time data analysis**.
- Improved efficiency and reduced operational costs.
- *Better understanding* of **system behavior** and **performance**.
- Increased collaboration between teams and stakeholders.



Challenges of Digital Twins

- Data **integration** and **management**.
- Ensuring data security and privacy.
- Maintaining the accuracy of the digital twin.
- Managing the complexity of the physical system.
- Ensuring interoperability between different systems and platforms.
- Scalability to accommodate large-scale systems.
- Cost of implementation and maintenance.
- User training and adoption.
- Regulatory compliance and standardization.
- Ethical considerations in data usage and analysis.
- Environmental impact of digital twin technologies.



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Thanks!

Questions?



Repo: <https://github.com/EngAndres/ud-public/tree/main/courses/systems-analysis>

